

How Our Track Feature Data Is Used

A quick guide for the team

1) What this dataset is

Each line represents a geotagged feature along the rail corridor. We use these points as reference landmarks to organize sensor findings and to visualize where potential issues are.

Record format (human readable)

- **K** *lat lon milepost code { JSON properties }* *

- **lat, lon:** WGS84 coordinates (use null if missing).
- **milepost:** decimal mile along the line (nullable).
- **code:** feature type enum.
- **properties:** JSON attributes like *name*, *crossing*, *start*, *end*, or *offset*.

Feature codes (used for styling & logic)

- **X** – Grade crossings (public/private). Often include an FRA ID like “844301C”.
- **MP** – Milepost markers, used to calibrate distance and align sensor data.
- **O** – Other assets or natural features (bridges, rivers, ponds).
- **S** – Segment annotations (e.g., curves) with optional *offset*.
- **CRF** – Corridor/region features with *start/end* mileposts.

2) How we use it

- **Mapping:** Points are exported to GeoJSON and loaded into GIS (QGIS, Mapbox, ArcGIS). Layers are styled by *code*, labeled by *name* or *milepost*.
- **Sensor alignment:** We join raw sensor readings (by GPS or mile marker) to the nearest feature. This adds context—e.g., a vibration spike near a bridge or a crossing.
- **QA & validation:** Check that mileposts increase along the corridor, that crossings have valid FRA IDs, and that segment/corridor features have required fields.
- **Alerting & trends:** Aggregate anomalies by feature type (e.g., all “X” crossings) and by mile to surface hotspots over time.
- **Maintenance workflow:** Flagged features become work orders with map links, photos, and last seen sensor values.

3) Minimal schema to keep things clean

Recommended columns:

- **lat** (float), **lon** (float), **milepost** (float/null), **code** (X/MP/O/S/CRF)
- **name** (str), **crossing** (str), **offset** (float), **start** (float), **end** (float)
- Use null for missing fields (avoid “-”).

Basic quality checks:

- Mileposts monotonic along the route; flag reversals.
- If *code* == *X*, *crossing* looks like a valid FRA ID.
- If *code* == *CRF*, both *start* and *end* present.
- If *code* == *S*, consider requiring *offset*.

4) Example (from a recent export)

Code	Count
X	30
MP	16
O	14
S	2
CRF	1

5) Handoff ready outputs

- Master **CSV** for easy editing/versioning.
- **GeoJSON** for map apps; one point per feature with all properties.
- Optional: a separate LineString for the track centerline; use it to snap points and autofill missing mileposts.

Next steps for the team

- Adopt the schema and checks above.
- Load the GeoJSON in QGIS/Mapbox and style by *code*.
- Connect sensor pipelines to these features for context and alerting.